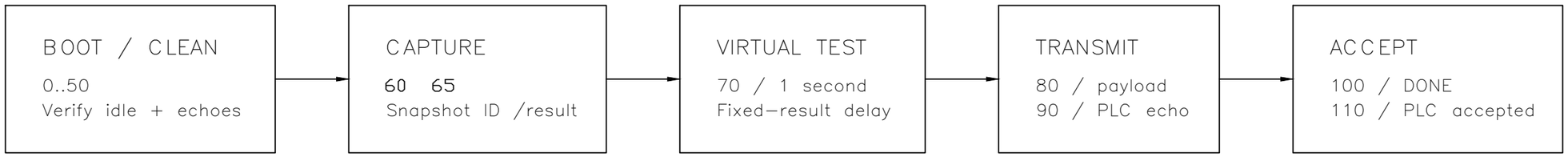

SIMULATED VOLTAGE MODE

Input voltage / low / high: I32 mV in0..5000.
Require low < high; limits inclusive.
PASS1: low <= voltage <= high; otherwise FAIL2.
Example defaults3300 /3000 /3600 mV.
Manual result mode can select PASS1/FAIL2/RETEST3.

SNAPSHOT AND FRESHNESS

Capture ID/result/measurement fields together at65.
ReadValid: no read error AND array length14.
D00030 changing confirms upstream PLC liveliness.
Unexpected alarm/state/request withdrawal stops the VI.
No real DUT source / DAQ / analog acquisition is present.

CSV RECORD CONTRACT

Columns: captured_utc, request_id, mode, voltage_mV, low_mV, high_mV, result_code, accepted_utc, accepted_id.
Write after accepted ID/result and PLC DONE echo1/request0; manual mode has three empty measurement fields.
CSV is inspection acceptance history. TOTAL is product completion at LS100 → 110: different events.
A product aborted after acceptance can legitimately have a CSV record without a completed-product count.

CSV append is not a crash-proof exactly-once database transaction. No TDMS, login audit trail or real measurement interface is claimed.

Keep single ownership of ATE result/DONE/response ID. Manual register tests and the automatic VI must not run concurrently.

NO	DATE	REVISION
R0	2026-10-22	FUNCTION RECORD

DWG. TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

LABVIEW VIRTUAL ATE AND ACCEPTANCE CSV

DRAWN BY	DATE
J. H. KO	2026-10-14

DWG. NO
009

SHEET NO
1 OF 1